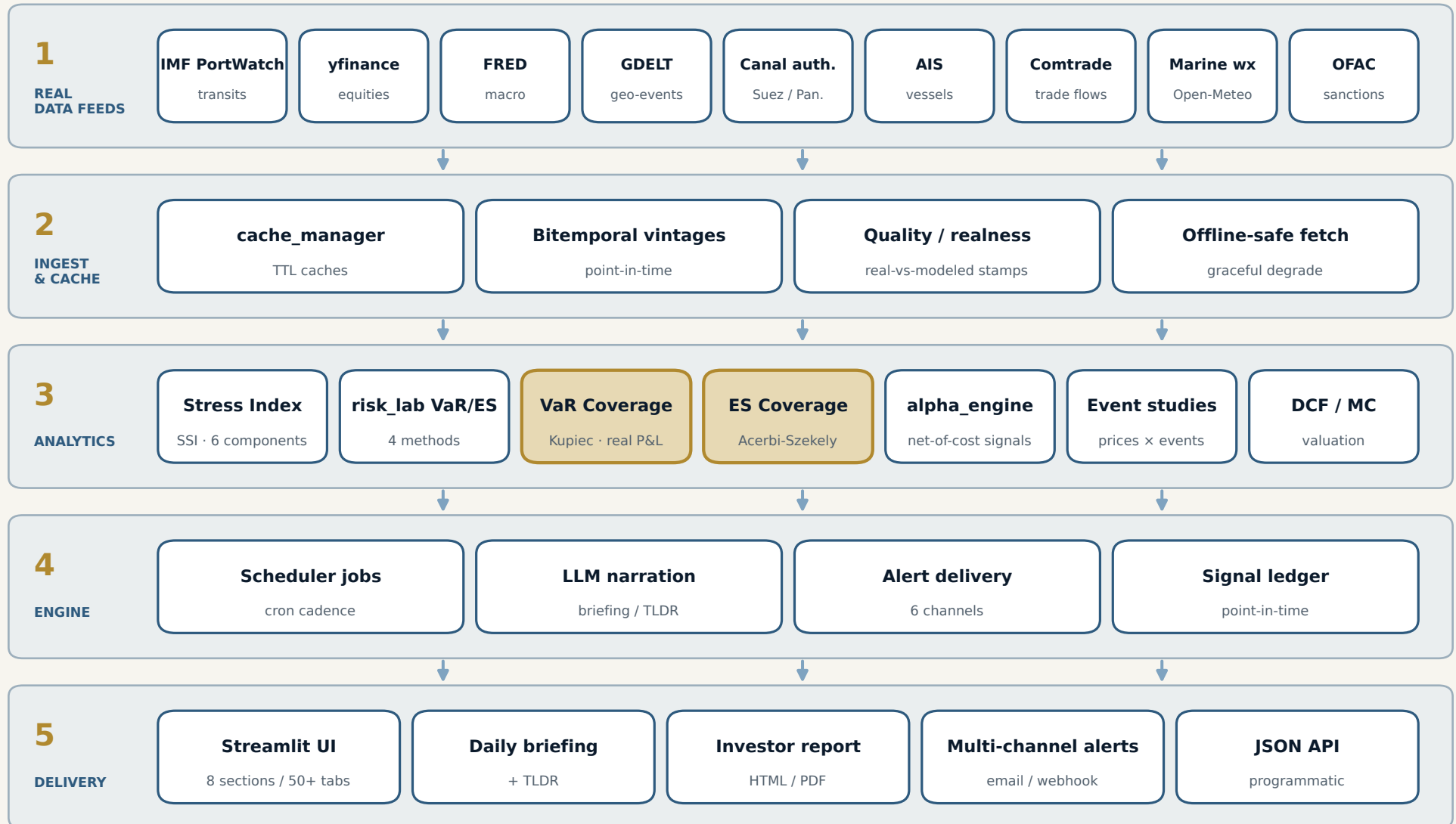


THE STACK

End-to-end, real to delivery

Real-data feeds flow down through ingest, analytics and the engine to delivery — labelled real vs modeled at every step.

 **BRASS** = the 2 validators scored on **REAL** market P&L (VaR Coverage · Kupiec; ES Coverage · Acerbi-Szekely)



20 backtest validators in all; the 2 brass nodes are scored on the real cache (1,292 trading days × 6 shipping equities). The other 18 are deterministic walk-forward / structural tests.